

DSA - Coding

CPP fundamental

```
#include <bits/stdc++.h>
using namespace std;

int main(){

    ios_base::sync_with_stdio(false);
    cin.tie(NULL);

    return 0;
}
```

← Standard template
(has a lot of
inbuilt maths
funⁿ.)

Description

Simple Calculator

Time Limit: 1 sec Memory: 256 MB

Description

Given two numbers X and Y . Print the summation, **multiplication**, and **subtraction** of these 2 numbers.

Input Format

Only one line containing two numbers X and Y separated by a space.

Output Format

Print 3 lines that contain the following in the same order:
First, print $\$x + y =$ followed by the summation result.
Second, print $\$x * y =$ followed by the multiplication result.
Third, print $\$x - y =$ followed by the subtraction result.

Constraints

$1 \leq X, Y \leq 10^5$

C++17 00:00:00 14 px

```
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5
6     ios_base::sync_with_stdio(false);
7     cin.tie(NULL);
8
9     long long a,b;
10
11     cin >> a >> b;
12
13     long long sum = a + b;
14     long long multi = a*b;
15     long long sub = a-b;
16
17     cout << a << " + " << b << " = " << sum << "\n";
18     cout << a << " * " << b << " = " << multi << "\n";
19     cout << a << " - " << b << " = " << sub << "\n";
20
21     return 0 ;
22 }
```

#mistakes

not followed
here

C++17 00:00:00 14 px

```
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5
6     ios_base::sync_with_stdio(false);
7     cin.tie(NULL);
8
9     int a,b;
10
11     int sum = a + b;
12     int multi = a*b;
13     int sub = a-b;
14
15     cin >> a >> b;
16
17     cout << sum << "\n";
18     cout << multi << "\n";
19     cout << sub << "\n";
20
21     return 0 ;
22 }
```

not proper
output

D. Difference

time limit per test: 1 second
memory limit per test: 256 megabytes

Given four numbers A, B, C and D . Print the result of the following equation :

$$X = (A * B) - (C * D).$$

Input

Only one line containing 4 separated numbers A, B, C and D ($-10^5 \leq A, B, C, D \leq 10^5$).

Output

Print "Difference = " without quotes followed by the equation result.

Examples

input

1 2 3 4

output

Difference = -10

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main() {
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long A, B, C, D;
9
10    cin >> A >> B >> C >> D;
11
12    long long X = (A*B)-(C*D);
13
14    cout << "Difference = " << X ;
15
16    return 0;
17 }
```

memory limit per test: 256 megabytes

Given a number R calculate the area of a circle using the following formula:

$$\text{Area} = \pi * R^2.$$

Note: consider $\pi = 3.141592653$.

Input

Only one line containing the number R ($1 \leq R \leq 100$).

Output

Print the calculated area, with 9 digits after the decimal point.

Example

input

2.00

output

12.566370612

Note

* Use the data type double for this problem.

** Use `setprecision(9)` to print 9 digits after decimal point.

*** you can use function `setprecision` that are in `#include<iomanip>` library for Example :

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    cout << fixed << setprecision(9);
    // your code.
}
```

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     cout << fixed << setprecision(9) ;
9
10    double R ;
11    double pi = 3.141592653;
12    cin >> R ;
13
14    double Radius = pi*R*R;
15
16    cout << Radius ;
17
18 }
```

this works here

#mistakes

```
1 #include <bits/stdc++.h>;
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     double R ;
9     double pi = 3.141592653;
10    cin >> R ;
11
12    double Radius = pi*R*R;
13
14    cout << Radius << setprecision(9) ;
15
16
17 }
```

no effect here.
calculation issue

this has to go before the main code.

G. Summation from 1 to N

time limit per test: 0.25 seconds
memory limit per test: 256 megabytes

Given a number N . Print the **summation** of the numbers that is between 1 and N (inclusive).

$$\sum_{i=1}^N i$$

Input

Only one line containing a number N ($1 \leq N \leq 10^9$)

Output

Print the **summation** of the numbers that are between 1 and N (inclusive).

Examples

input	output
3	6
10	55

Precedence	Operator	Description	Associativity
1 (Highest)	()	Grouping (Parentheses)	Left-to-right
2	+ -	Unary Plus / Minus	Right-to-left
3	* / %	Multiplication, Division, Modulo	Left-to-right
4 (Lowest)	+ -	Addition, Subtraction	Left-to-right

$$Sol^n = 0.5 * n(2*1 + (n-1)*d)$$



```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long N, d, a;
9     long long Final;
10
11     d = 1;
12     a = 1;
13
14     cin >> N;
15
16     Final = ((N)*(N+1))/2;
17
18     cout << Final ;
19
20     return 0;
21 }
```

$a = 1$
 $d = 1$
 $N = n$

$$AP = S_n = \frac{n(2a + (n-1)d)}{2}$$

#Mistake

#Mistake

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     int N, d, a;
9     double Final;
10
11     d = 1;
12     a = 1;
13
14     cin >> N;
15
16     Final = ((N/2) * (2*a + (N-1)*d));
17
18     cout << Final ;
19
20     return 0;
21 }
```

integer overflow

increases very high in some cases

```

1 #include <bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     int N, d, a;
9     double Final;
10
11     d = 1;
12     a = 1;
13
14     cin >> N;
15
16     Final = ((N)*(2*a + (N-1)*d))/2;
17
18     cout << Final ;
19
20     return 0;
21 }
```

time limit per test: 0.25 seconds
memory limit per test: 64 megabytes

Given two numbers N and M . Print the **summation** of their **last digits**.

Input

Only one line containing two numbers N, M ($0 \leq N, M \leq 10^{18}$).

Output

Print the answer of the problem.

Example

input	Copy
13 12	
output	Copy
5	

Note

First Example :

last digit in the first number is **3** and **last digit** in the second number is **2**.

So the answer is: $(3 + 2 = 5)$

#Mistake

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long M, N;
9     long long m, n;
10    long long Ans ;
11
12    cin >> M, N;
13    m = M % 10 ;
14    n = N % 10 ;
15
16    Ans = m + n ;
17
18    cout << Ans;
19
20    return 0;
21
22 }
```

not correct way to separate

```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long M, N;
9     long long m, n;
10    long long Ans ;
11
12    cin >> M >> N;
13    m = M % 10 ;
14    n = N % 10 ;
15
16    Ans = m + n ;
17
18    cout << Ans;
19
20    return 0;
21
22 }
```

correct way

time limit per test: 1 second
memory limit per test: 256 megabytes

Given 2 numbers A and B . Print **floor**, **ceil** and **round** of A/B

Input

Only one line containing two numbers A and B ($1 \leq A, B \leq 10^3$)

Output

Print 3 lines that contain the following in the same order:

1. "floor A/B = **Floor result**" without quotes.
2. "ceil A/B = **Ceil result**" without quotes.
3. "round A/B = **Round result**" without quotes.

Examples

input	Copy
10 3	
output	Copy
floor 10 / 3 = 3 ceil 10 / 3 = 4 round 10 / 3 = 3	
input	Copy
10 4	
output	Copy
floor 10 / 4 = 2 ceil 10 / 4 = 3 round 10 / 4 = 3	

```
if (round > 0.4)
    round = ceil;
else round = floor;
```

not req. here
imp syntax
flawed logic

floor
ceil
round
all
inbuilt
maths
fun

3 → floor = 3
3 → ceil = 3
fails

#Mistake

```
1 #include<bits/stdc++.h>
2 using namespace std ;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long A, B;
9     cin >> A >> B ;
10    long double op = A/B;
11
12    long long floor = op/1 ;
13    long long ceil = floor + 1 ;
14
15    long double round = op - floor ;
16    if (round > 0.4)
17        round = ceil;
18    else round = floor;
19
20    cout << "floor " << A << "/" << B << " = " << floor
21    << endl ;
22    cout << "ceil " << A << "/" << B << " = " << ceil <<
23    endl ;
24    cout << "round " << A << "/" << B << " = " << round
25    << endl ;
26
27    return (0);
28 }
```

```
1 #include<bits/stdc++.h>
2 using namespace std ;
3
4 int main(){
5     ios::sync_with_stdio(false);
6     cin.tie(nullptr);
7
8     long long A, B;
9     cin >> A >> B ;
10    long double op = (long double)A/B;
11
12    cout << "floor " << A << " / " << B << " = " << floor
13    (op) << endl ;
14    cout << "ceil " << A << " / " << B << " = " << ceil
15    (op) << endl ;
16    cout << "round " << A << " / " << B << " = " << round
17    (op) << endl ;
18
19    return (0);
20 }
```

type casting
to avoid this

Trap 1: Integer Division (Line 10): You declared `op` as a `long double`, which is good. But `A` and `B` are still integers. When you write `A/B`, C++ does integer division *first*. For `122 / 16`, it calculates 7, throws away the decimal completely, and *then* stores `7.0` in